Interpretable Machine Learning for Emergency Department Triage: A Systematic Literature Review

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***Abstract*— The triage system determines the severity of a patient's condition and the priority of emergency care. Conventional methods are still limited by the subjectivity of medical personnel, which leads to inconsistent decisions. Machine learning (ML) models improve the accuracy of clinical condition classification, although they still have limitations in terms of lack of readability, making prediction results difficult for medical personnel to understand. Interpretable machine learning (IML) approaches can overcome this problem without reducing prediction performance in decision-making. This systematic review analyzes 25 studies published between 2021 and 2025 in five major databases: Scopus, IEEE Xplore, ACM Digital Library, PubMed, and SpringerLink. The studies were analyzed based on their focus on research context analysis, dataset type, modeling methods, interpretability techniques, and reported final results. Two approaches dominated: gradient boosting models using SHAP or LIME, and score-based systems such as AutoScore and SERP. The reported prediction outcomes included hospital admissions, intensive care needs, mortality, risk of sepsis, and length of patient stay. Existing research gaps include a lack of multimodal methods, evaluation of fairness constraints, and a lack of data sources from low- and middle-income countries.**

Keywords—Interpretable machine learning, emergency department, triage, electronic health records, systematic literature review.

I. INTRODUCTION

A. Conventional Triage System

The triage procedure in the emergency department (ED) is performed to determine the clinical urgency and classify the patient's condition. The triage system has several standards that are widely used around the world, such as the Australasian Triage Scale (ATS), Emergency Severity Index (ESI), Canadian Triage and Acuity Scale (CTAS), and Manchester Triage System (MTS). However, conventional triage has several limitations, such as high subjectivity, which increases the risk of under-triage and over-triage. This risk occurs when critical patients wait too long, while patients with minor conditions receive high priority. Such issues can lead to inefficiencies of services and resource allocation in health facilities.

B. From Machine Learning to Interpretable Machine Learning

Machine learning (ML) is considered a promising alternative approach to improving triage accuracy. Empirical evidence from recent studies show that ML models outperform conventional triage systems [1]-[3]. However, most ML algorithms are regarded as black boxes because their predictions are not accompanied by transparent reasoning. This limitation reduces the level of trust among medical professionals and may hinder clinical adoption.

Interpretable machine learning (IML) was developed to overcome this limitation. Several inherently interpretable models, such as those based on logistic regression and decision trees, can be directly understood by users [4], [5]. In another approach, complex models can be explained through post-hoc interpretability methods such as SHAP and LIME, which describe the contribution of each feature to the prediction [2], [6]. By integrating algorithmic reasoning with clinical intuition, IML plays an important role in enhancing trust and acceptance among medical personnel, thereby promoting the adoption of AI-based decision support in emergency care.

C. Objectives

The objectives of this SLR are: (1) summarizing the latest innovations in the application of IML in triage systems in EDs, (2) mapping methodological gaps found in previous studies and challenges in their implementation, and (3) recommending future research directions that emphasize model performance balance, transparency, and opportunities for integration in clinical settings. Through this review, we aim to contribute by outlining methodological advances and practical implications that support digital transformation in healthcare, in this case in emergency triage systems. Based on this framework, we pose the following research questions: how has IML developed in ED triage between 2021 and 2025, and what methodological gaps have been identified that could be opportunities for implementation and future challenges that still need to be addressed?

II. METHODOLOGY

We used PRISMA 2020 as a guideline for the methodology used in conducting this review.

A. PICOC

The PICOC framework is used as a reference in defining the scope of the study. This approach ensures that every aspect we analyze remains consistent with the core objectives of the study. Each component of the PICOC is described to establish the boundaries and limitations for conducting the analysis.

1) Population (P) : Patients of all ages who come to the ED and undergo initial triage.
2) Intervention (I) : All IML models, inherently interpretable models, and complex models.
3) Comparison (C) : Conventional triage scale or ML model that cannot be interpreted.
4) Outcomes (O) : Predictive accuracy, interpretability level, and operational outcomes.

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5) Context (C) : Triage in the ED that utilize electronic health record (EHR) data, manual triage forms or multimodal data.

B. Search Strategy

The literature search mechanism in this review was conducted in a structured manner across several databases, particularly those related to medical research. The databases referenced were Scopus, IEEE Xplore, PubMed, ScienceDirect, and Springer Nature.

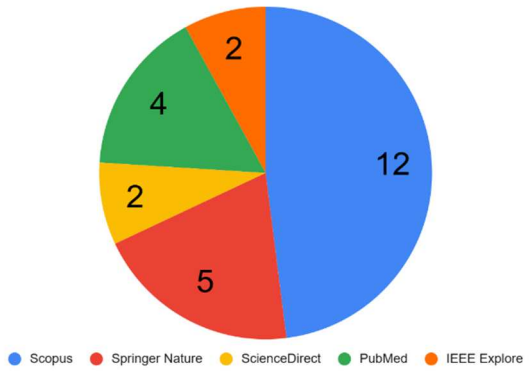


Fig. 1. Distribution of Sources of Articles

We developed a search method using a combination of keywords with Boolean operators, with main keywords such as "emergency department," "triage," and "interpretable machine learning." We limited the publication time frame to the period from 2021 to 2025. The distribution of search results based on the reference database is presented in Fig 1. Meanwhile, the article selection process is presented in Section 3.

C. Inclusion and Exclusion Criteria

The determination of inclusion and exclusion criteria is explained in detail in Table I.

TABLE I. INCLUSION AND EXCLUSION CRITERIA

Category	Criteria
Inclusion	• Journal articles have been published between 2021 and 2025. • Journal articles have undergone peer review or conference review. • The content of the research article states that IML has been applied to triage tasks in the ED. • Quantitative performance metrics are reported in the article. • The research in the article uses structured triage data or unstructured data.
Exclusion	• Studies that use black-box machine learning models without interpretability. • Studies conducted outside the context of the ED. • Articles in the form of reviews, editorials, preprints, or literature in the gray category.

D. Study Selection and Data Extraction

The article selection process begins by filtering search results based on title and abstract. Then, for articles that pass the initial stage, we conduct a thorough review to ensure their relevance. The review was conducted jointly by both authors to ensure consistency. The selected articles that passed this stage were then included in the final analysis.

The next step in this stage was to extract details of the publication, dataset characteristics, machine learning methods and interpretability techniques applied, as well as the type of prediction task and the performance metrics analyzed. The extracted information was then summarized and presented in a comprehensive table in the Results section, as shown in Table II.

TABLE II. QUALITY ASSESSMENT CRITERIA

NO.	Quality Assessment Criteria
1	Focus only on triage tasks in the ED.
2	The model used is interpretable.
3	Enables replication of the methodology, which is clearly explained.
4	Assesses hospital efficiency and related operational metrics.
5	Results are presented transparently, including strengths and limitations.
6	Uses real-world datasets from EDs or hospitals.
7	Discusses challenges faced in implementing the model in clinical practice.
8	Has been published through a peer-review process.
9	Compares with traditional triage systems or machine learning models that are still black-box.
10	Patient clinical outcomes are used as key indicators.
11	Statistical significance is adequate to support findings.
12	Consistency between conclusions and analysis results.

E. Quality Assessment

At this stage, we adapted the MINIMAR framework as evidence-based reporting for machine learning [12] and also the CONSORT-AI/SPIRIT-AI principles for artificial intelligence-based interventions in healthcare [13]. This aims to ensure that each study analyzed meets comprehensive methodological standards by creating a 12-point checklist covering aspects such as clinical relevance, reporting transparency, use of real-world data, and challenges in implementation.

We evaluate each study based on these 12 criteria, with a score of 1 given if the criteria are met and a score of 0 if they are not. Studies with scores above 7 are categorized as high quality, scores between 5 and 7 as medium, and below 5 as low.

III. RESULT

A. Study Selection

We obtained a total of 25 studies after going through four stages of selection. In the first stage, we obtained 362 publications from a literature search. Second, 62 articles were eliminated because they were identified as duplicates or did not meet the criteria. From the remaining 300, the third stage was carried out by examining the relevance of the titles and abstracts to the topic of IML in triage in the ED. Sixty articles were found to be fully evaluated in the fourth stage, of which 35 articles were excluded because they did not meet the inclusion criteria. The selection stage flow is presented in Fig. 2 in the form of a PRISMA diagram.

B. Descriptive Statistics

The analysis found that most studies used data sourced from high-income countries such as the United States, South Korea, Singapore and China. Meanwhile, data sources from LMICs were still very limited, as shown in Fig. 3.

Other findings from this study indicate that predictions of patient mortality, likelihood of hospitalization, length of stay (LOS) and need for intensive care emerged as the most dominant research themes, as shown in Fig. 4. Meanwhile, a small portion of the remaining studies focused on predicting

sepsis and patient waiting time. Another point we highlight is that the type of data used in most studies still utilizes structured data obtained from electronic health records (EHR) or manual triage forms, as illustrated in Fig. 5.

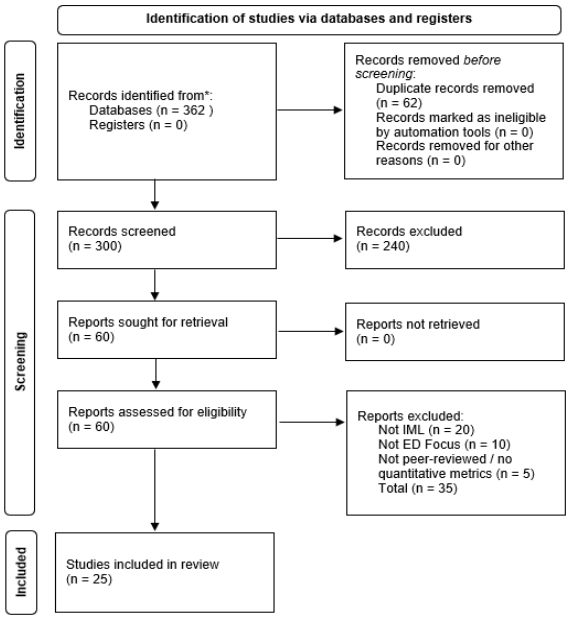


Fig. 2. PRISMA 2020 flow diagram of study selection process

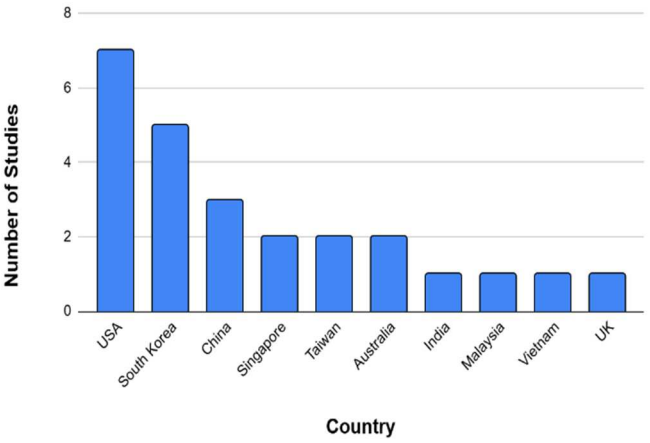


Fig. 3. Country Distribution of Included Studies

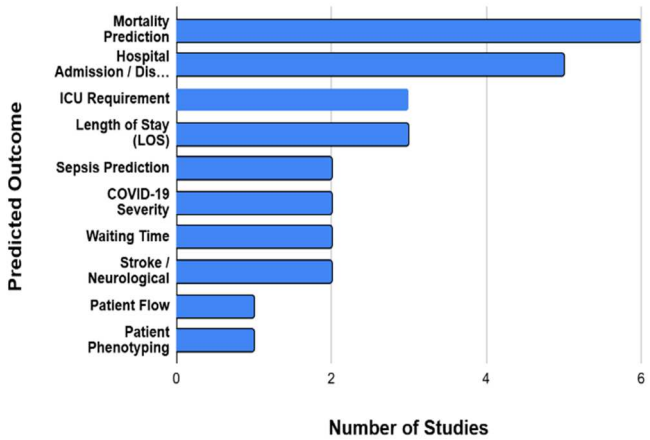


Fig. 4. Outcome Distribution of Included Studies

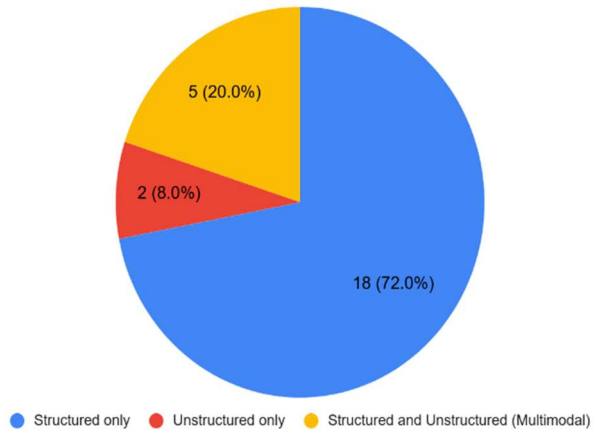


Fig. 5. Triage Data Type Distribution of Included Studies

Gradient boosting methods such as XGBoost, LightGBM, and CatBoost are the most widely used methods. These methods are usually accompanied by SHAP or LIME, which serve as their interpretation. A summary of the study characteristics is presented in Table III.

C. Performance Overview

Most studies use AUROC as the main indicator for evaluating models. In addition, other metrics such as sensitivity, specificity, and calibration are also taken into account. Our analysis found a fairly high level of prediction accuracy, with reported AUROC values ranging from 0.78 to 0.96. The highest AUROC value was seen in a study conducted by Baek et al., on the application of XGBoost with SHAP to predict hospitalization [2]. Meanwhile, the use of simple decision trees in the Xie et al., study [4], which was used to estimate mortality, had the lowest value, although it was still in the good accuracy category. The AutoScore approach in the study by Gebran et al. [15] achieved an AUROC value of 0.88 and was considered superior in terms of interpretability. Higher AUROC values were achieved through deep learning approaches that emphasize explainability [17], [18], where the figure reached 0.82.

Several studies also evaluated other aspects such as calibration [19] and fairness [11], [12]. performance scores obtained were quite high, there was still variation among patient groups. The distribution of AUROC values from all these studies can be seen in Fig. 6.

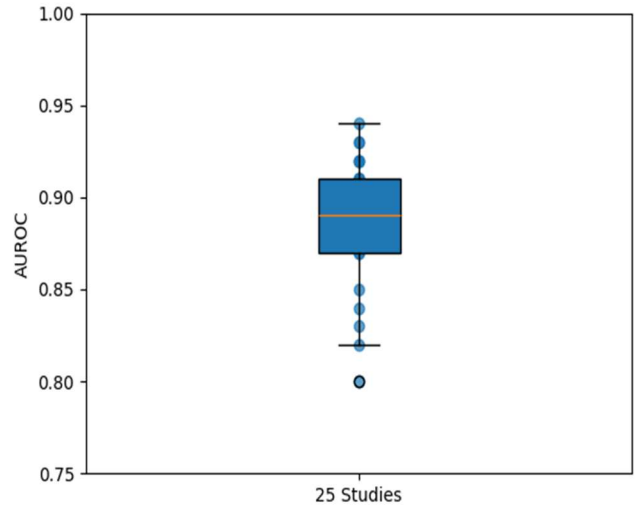


Fig. 6. Distribution of AUROC for IML Models in ED Triage Studies

TABLE III. A SUMMARY OF STUDY CHARACTERISTICS

No	Ref	Data Source	Data Type	Predicted Outcome(s)			Model(s) Used	Interpretability	Auroc Score	
				Hospital Admission	*ICU Needs*	*Other(s)*			*ML*	*Convensional*
1	[1]	UK and EU ED multicenter	Structured	Yes	No	Patient Disposition	Gaussian Process ML	Shapley-based Uncertainty Quantification	0.88 - 0.93	Logistic Regression = 0.84
2	[2]	South Korea (COVID-19 cohort)	Structured	Yes	Yes	Mortality	XGBoost, LightGBM	SHAP (global and local)	0.83 - 0.94	CURB-65 = 0.76
3	[3]	Singapore (multicenter)	Structured	Yes	No	Mortality	AutoScore (rule-based ML)	Inherent transparency and SHAP	0.84 - 0.89	ESI = 0.81
4	[4]	Singapore General Hospital	Structured	No	No	Mortality (2-30 days)	AutoScore adapted as SERP	Rule-based interpretability	0.82	ESI = 0.77
5	[5]	Pan-Asian Trauma Registry	Structured	No	Yes	Mortality	AutoScore adapted as GIFT	Rule-based scoring	0.76 - 0.85	RTS/TRISS = 0.73
6	[6]	China (tertiary hospital)	Structured	No	Yes	Sepsis Risk	XGBoost, LightGBM	SHAP	0.91 - 0.94	Logistic Regression = 0.85
7	[7]	UK NHS amd Australia ED	Structured	Yes	No	LOS, Disposition	RF, XGBoost	SHAP, visualization	0.80 - 0.87	ESI = 0.78
8	[8]	China (ICU dataset)	Structured	No	Yes	ICU Admission	GBM, RF	SHAP and Decision Rule	0.73 - 0.80	Logistic Regression = 0.71
9	[9]	Cyprus (MIMIC-IV-ED)	Structured	No	Yes	ICU Mortality	GB, RF, MLP	SHAP (meta-ensemble)	0.72 - 0.73	Logistic Regression = 0.67
10	[10]	Cyprus (multi-hospital)	Structured	No	No	ED LOS	GB and Rule Extraction	Custom Rule-based	0.73	Not reported
11	[11]	USA (JPS Health Network)	Structured	No	No	Wait Time	XGBoost, RF, ANN	SHAP and PDP	0.79	Baseline Regression = 0.70
12	[12]	USA (Multi-hospital)	Structured	No	No	Fairness Evaluation	XGBoost	SHAP (fairness)	0.79	Not reported
13	[15]	USA (MGH)	Structured	No	Yes	ICU Need	LR and GBM Hybrid	SHAP	0.88	Clinician Estimate = 0.81
14	[16]	USA (Yale and MIMIC)	Structured	Yes	No	Patient Phenotyping	Auto-encoder and k-Means	Visualization (UMAP and Heatmap)	Not report-ed	Not reported
15	[17]	Taiwan (NTU Hospital)	Structured and un-structured	Yes	No	LOS, Triage Level	TabNet and MacBERT	Attention Map and Feature Weights	0.82	CNN baseline = 0.74
16	[18]	USA (PECARN)	Structured	No	No	TBI Identification	DNN, RF, SVM	Rule Extraction (PermFIT)	0.79 - 0.91	Logistic Regression = 0.80
17	[19]	Ireland (mental health ED)	Structured	Yes	No	Admission (Psychiatric)	XGBoost and LR	SHAP and Coefficient Analysis	0.84	Logistic Regression = 0.72
18	[20]	Greece (Regional ED)	Structured	Yes	No	Disposition	RF, XGBoost	SHAP (feature importance)	0.91	Manual Rules = 0.82
19	[21]	UK (Oxford Hospitals)	Structured	Yes	Yes	Admission Location dan Flow	Deep Interpret-able Network	LRP	0.87	Logistic Regression = 0.81
20	[22]	Singapore (SGH ED)	Structured	No	Yes	Mortality (2-30 days)	AutoScore and Imbalance Framework	Feature scoring	0.87 - 0.91	SERP = 0.83
21	[23]	USA (Hospital ED flow data)	Structured	Yes	No	Patient Flow	CNN and LSTM	Activation Mapping and SHAP	0.86	Rule-Based System = 0.75
22	[24]	India (Multi-hospital)	Structured	Yes	Yes	Mortality	XGBoost, RF, LR	SHAP and LIME	0.85	SOFA Score = 0.78
23	[25]	Asia (multicenter ED)	Structured	Yes	No	Triage Score Validation	AutoScore (SERP ext.)	Score-based transparency	0.88 - 0.92	Manual Triage = 0.84
24	[26]	South Korea (SNUH)	Structured	No	Yes	Critical Care Outcome	XGBoost, DNN	PDP and Global Interpretability	0.86	KTAS = 0.79
25	[27]	China (Multi-center)	Structured	Yes	No	Ischemic Stroke Risk	XGBoost, RF, DNN	SHAP and LIME	0.91 - 0.92	Cox Model = 0.80

D. Synthesis

Our literature synthesis identified two main approaches in the application of interpretable machine learning used in triage systems in ED's. The first approach makes extensive use of gradient boosting models such as XGBoost and LightGBM, which are generally combined with post-training interpretability methods such as SHAP or LIME, demonstrating strong predictive performance and high levels of interpretability [2], [6], [11], [12], [20], [24], [27]. These models are considered relevant as tools to support clinical decision-making [6], [7], [9], [19]. The second approach uses inherently transparent models, such as AutoScore and decision trees [3]-[5], [22], [25]. Although the second model shows less accurate performance, it is still considered feasible because it can still explain the resulting clinical decisions [4], [5], [22].

Based on these two proven approaches, recent studies also show continuous innovation in modeling techniques and how prediction results are interpreted. In addition to commonly used classical methods such as SHAP and LIME, several studies have also explored other visualization-based and attention mechanism-based interpretation methods [7], [9], [16], [17], [18]. Another finding is the development of multimodal frameworks that integrate structured and unstructured triage data [16]-[18]. These efforts aim to bridge the gap between computational accuracy and clinical assessments derived from medical professionals' experience. This is done to ensure that AI-based predictions are not only acceptable, but also fully trusted and verified in real-world emergency care settings [6], [7], [16], [17].

Along with the progress made, various major challenges such as methodological and implementation aspects were found in this field of research. Most studies focus on clinical prediction outcomes including mortality, hospitalization, intensive care needs, and sepsis [2]-[9]. Meanwhile, studies that include predictive outcomes that can support operational aspects such as waiting times, patient flow, and resource use efficiency have received relatively little attention [7], [10]-[12]. Another challenging aspect is related to model fairness and generalization [11], [12]. This is because the data sources used do not yet represent data from LMICs and there is inconsistency in model performance among patient groups with different features and risk profiles [7], [9], [10], [12], [19]. We also found a high dependence on structured EHRs [2]-[4], [6], [7], [10], [11], [20]. This limits the utilization of richer clinical contexts found in narrative medical records. Although some early studies have explored multimodal architectures, but their implementation remains limited in both scale and empirical validation [16]-[18].

E. Limitations

Although this study has many strengths, this review still has several limitations. First, we only selected English-language publications, which may introduce language bias. Second, most of the datasets analyzed were from high-income countries, limiting the generalizability of the findings, especially for LMICs. These limitations complement the data representation challenges identified in Section D. Finally, publication bias and incomplete reporting in some studies may also affect the synthesis results.

F. Future Work

Research in this field remains open for further exploration. Researcher can expand the scope of research that not only focuses on how to predict clinical outcomes, but also considers integrating operational metrics such as patient waiting times and optimizing the workload of medical staff in healthcare facilities. In addition, in the future, researchers can try to design models to optimize resource efficiency while applying fairness metrics for diverse patient populations, as summarized in Table IV.

TABLE IV. FUTURE WORK

Research Topic	Identified Gap	Future Direction	Potential Contribution
Outcome evaluation	Operational outcomes rarely evaluated	Include waiting time and efficiency metrics	Improved resource allocation and patient flow
Data diversity	Limited multimodal integration	Combine structured and unstructured triage data	Richer and more realistic clinical prediction
Fairness and generaliz-ability	Bias across subgroups, under-representation of LMIC	Apply fairness metrics, expand LMIC datasets	More equitable and globally relevant models
Validation and deployment	Lack of prospective and multicenter validation	Conduct real-world and multicenter implementations	Evidence for adoption and scalability in EDs

IV. CONCLUSION

Through the SLR process, we have identified and analyzed a number of findings. Over the past five years, there has been significant progress in the use of IML in triage in EDs. The approach used is dominated by gradient boosting models, most of which are implemented in conjunction with SHAP or LIME. However, several other methods are also taken into account because the accuracy performance they produce is in the high category.

However, there are still several challenges, including the fact that most studies focus only on clinical outcomes such as mortality, hospitalization, intensive care, and sepsis, while operational outcomes such as patient waiting times and resource efficiency are less evaluated. Additionally, issues of fairness that result in performance variations between subpopulations and generalizations due to the lack of datasets from LMICs remain crucial issues. In conclusion, we believe that IML has great potential to drive digital transformation in the ED triage system. Future research should not only focus on prediction accuracy but also emphasize fairness, operational relevance, multimodal data integration, and application in resource-constrained hospitals. This is necessary to produce models that are not only accurate but also transparent, fair, and clinically acceptable.

Although this review is centered on triage systems in the ED, we hope that the methodological insights derived from this study may contribute to the broader application of IML in in decision support, both for clinical and operational purposes across healthcare systems.

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REFERENCES

[1] A. S. B. Abdulai, J. Storm and M. Ehrlich, "'I don't know': An uncertainty-aware machine learning model for predicting patient disposition at emergency department triage," Int. J. Med. Inform., vol. 201, Art. no. 105957, Apr. 2025, doi: 10.1016/j.ijmedinf.2025.105957.

[2] S. Baek, Y. J. Jeong, Y.-H. Kim, J. Y. Kim, J. H. Kim, E. Y. Kim, J.-K. Lim, J. Kim, Z. Kim, K. Kim and M. J. Chung, "Development and validation of a robust and interpretable early triaging support system for patients hospitalized with COVID-19: Predictive algorithm modeling and interpretation study," J. Med. Internet Res., vol. 26, no. 1, Art. no. e52134, Jan. 2024, doi: 10.2196/52134.

[3] J. Y. Yu, F. Xie, L. Nan, S. Yoon, M. E. H. Ong, Y. Y. Ng and W. C. Cha, "An external validation study of the Score for Emergency Risk Prediction (SERP), an interpretable machine learning–based triage score for the emergency department," Sci. Rep., vol. 12, Art. no. 17466, Oct. 2022, doi: 10.1038/s41598-022-22233-w.

[4] F. Xie, M. E. H. Ong, J. N. M. H. Liew, K. B. K. Tan, A. F. W. Ho, G. D. Nadarajan, L. L. Low, Y. H. Kwan, B. A. Goldstein, D. B. Matchar, B. Chakraborty and N. Liu, "Development and assessment of an interpretable machine learning triage tool for estimating mortality after emergency admissions," JAMA Netw. Open, vol. 4, no. 8, Art. no. e2118467, Aug. 2021, doi: 10.1001/jamanetworkopen.2021.18467.

[5] J. Y. Yu, S. Heo, F. Xie, N. Liu, S. Y. Yoon, H. S. Chang, T. Kim, S. U. Lee, M. E. H. Ong, Y. Y. Ng, S. D. Shin, K. Kajino and W. C. Cha, "Development and Asian-wide validation of the Grade for Interpretable Field Triage (GIFT) for predicting mortality in pre-hospital patients using the Pan-Asian Trauma Outcomes Study," Lancet Reg. Health West. Pac., vol. 34, Art. no. 100733, Mar. 2023, doi: 10.1016/j.lanwpc.2023.100733.

[6] Z. Liu, W. Shu, T. Li, X. Zhang and W. Chong, "Interpretable machine learning for predicting sepsis risk in emergency triage patients," Sci. Rep., vol. 15, no. 887, Jan. 2025, doi: 10.1038/s41598-025-85121-z.

[7] L. Song, U. Aickelin, T. N. Fazio, A. Sharma, M. Kouhounestani, S. Plumb and M. J. Putland, "Developing interpretable machine learning models to predict length of stay and disposition decision for adult patients in emergency departments," BMJ Health Care Inform., vol. 32, Art. no. e101152, May 2025, doi: 10.1136/bmjhci-2024-101152.

[8] Z. Liu, W. Shu, H. Liu, X. Zhang and W. Chong, "Development and validation of interpretable machine learning models for triage patients admitted to the intensive care unit," PLoS One, vol. 20, no. 2, Art. no. e0317819, Feb. 2025, doi: 10.1371/journal.pone.0317819.

[9] H. Y. Başer, T. Evran and M. A. Cifci, "Machine learning-augmented triage for sepsis: Real-time ICU mortality prediction using SHAP-explained meta-ensemble models," Biomedicines, vol. 13, no. 6, Art. no. 1449, Jun. 2025, doi: 10.3390/biomedicines13061449.

[10] W. A. Sulaiman, C. Stylianides, A. Nikolaou, Z. Antoniou, I. Constantinou, L. Palazis, A. Vavlitou, T. Kyprianou, E. Kyriacou, A. Kakas, M. S. Pattichis, A. S. Panayides and C. S. Pattichis, "Leveraging machine learning and rule extraction for enhanced transparency in emergency department length of stay prediction," Front. Digit. Health, vol. 6, Art. no. 1498939, Feb. 2025, doi: 10.3389/fdgth.2024.1498939.

[11] H. Wang, N. Sambamoorthi, D. Sandlin and U. Sambamoorthi, "Interpretable machine learning models for prolonged emergency department wait time prediction," BMC Health Serv. Res., vol. 25, no. 403, Mar. 2025, doi: 10.1186/s12913-025-12535-w.

[12] H. Wang, N. Sambamoorthi, N. Hoot, D. Bryant and U. Sambamoorthi, "Evaluating fairness of machine learning prediction of prolonged wait times in emergency department with interpretable extreme gradient boosting," PLoS Digit. Health, vol. 4, no. 3, Art. no. e0000751, Mar. 2025, doi: 10.1371/journal.pdig.0000751.

[13] T. Hernandez-Boussard, S. Bozkurt, J. P. A. Ioannidis and N. H. Shah, "MINIMAR (Minimum information for medical AI reporting): Developing reporting standards for artificial intelligence in health care," Nat. Med., vol. 26, pp. 1320–1324, Aug. 2020, doi: 10.1038/s41591-020-1041-y.

[14] CONSORT-AI and SPIRIT-AI Working Group, "Reporting guidelines for clinical trial protocols and reports for interventions involving artificial intelligence: The SPIRIT-AI and CONSORT-AI guidelines," Nat. Med., vol. 26, pp. 1364–1374, Aug. 2020, doi: 10.1038/s41591-020-1037-7.

[15] A. Gebran, A. Vapsi, L. R. Maurer, M. El Moheb, L. Naar, S. S. Thakur, R. Sinyard, D. Daye, G. C. Velmahos, D. Bertsimas and H. M. A. Kaafarani, "POTTER-ICU: An artificial intelligence smartphone-accessible tool to predict the need for intensive care after emergency surgery," Surgery, vol. 172, no. 2, pp. 470–475, Apr. 2022, doi: 10.1016/j.surg.2022.03.023.

[16] N. C. Hurley, A. D. Haimovich, R. A. Taylor and B. J. Mortazavi, "Visualization of emergency department clinical data for interpretable patient phenotyping," Smart Health, vol. 25, Art. no. 100285, Apr. 2022, doi: 10.1016/j.smhl.2022.100285.

[17] Y.-T. Lin, Y.-X. Deng, C.-L. Tsai, C.-H. Huang and L.-C. Fu, "Interpretable deep learning system for identifying critical patients through the prediction of triage level, hospitalization, and length of stay: Prospective study," JMIR Med. Inform., vol. 12, Art. no. e48862, Jan. 2024, doi: 10.2196/48862.

[18] B. Zou, X. Mi, E. Stone and F. Zou, "A deep neural network framework to derive interpretable decision rules for accurate traumatic brain injury identification of infants," BMC Med. Inform. Decis. Mak., vol. 23, no. 58, Mar. 2023, doi: 10.1186/s12911-023-02155-x.

[19] O. Higgins, S. K. Chalup and R. L. Wilson, "Machine learning model reveals determinators for admission to acute mental health wards from emergency department presentations," Int. J. Ment. Health Nurs., vol. 33, no. 6, pp. 2354–2369, Aug. 2024, doi: 10.1111/inm.13402.

[20] G. Feretzakis, A. Sakagianni, A. Anastasiou, I. Kapogianni, R. Tsoni, C. Koufopoulou, D. Karapiperis, V. Kaldis, D. Kalles and V. S. Verykios, "Machine learning in medical triage: A predictive model for emergency department disposition," Appl. Sci., vol. 14, no. 15, Art. no. 6623, Jul. 2024, doi: 10.3390/app14156623.

[21] R. El-Bouri, D. W. Eyre, P. Watkinson, T. Zhu and D. A. Clifton, "Hospital admission location prediction via deep interpretable networks for the year-round improvement of emergency patient care," IEEE J. Biomed. Health Inform., vol. 25, no. 1, pp. 289–299, Jan. 2021, doi: 10.1109/JBHI.2020.2990309.

[22] C. S. J. Look, S. Teixayavong, T. Djärv, A. F. W. Ho, K. B. K. Tan and M. E. H. Ong, "Improved interpretable machine learning emergency department triage tool addressing class imbalance," Digit. Health, vol. 10, pp. 1–12, May 2024, doi: 10.1177/20552076241240910.

[23] A. R. Sharafat and M. Bayati, "PatientFlowNet: A deep learning approach to patient flow prediction in emergency departments," IEEE Access, vol. 9, pp. 45552–45563, Mar. 2021, doi: 10.1109/ACCESS.2021.306616.

[24] S. Bhatia, Y. Makhija, S. Jayaswal, S. Singh, P. S. Malik, S. K. Venigalla, P. Gupta, S. N. Samaga, R. N. Hota and I. Gupta, "Severity and mortality prediction models to triage Indian COVID-19 patients," PLoS Digit. Health, vol. 1, no. 3, Art. no. e0000020, Mar. 2022, doi: 10.1371/journal.pdig.0000020.

[25] J. Y. Yu, D. Kim, S. Yoon, T. Kim, S. Heo, H. Chang, G. S. Han, K. W. Jeong, R. W. Park, J. M. Gwon, F. Xie, M. E. H. Ong, Y. Y. Ng, H. J. Joo and W. C. Cha, "Inter-hospital external validation of interpretable machine learning-based triage score for the emergency department using common data model," Sci. Rep., vol. 14, Art. no. 6666, Apr. 2024, doi: 10.1038/s41598-024-54364-7.

[26] H. Yun, J. Choi and J. H. Park, "Prediction of critical care outcome for adult patients presenting to emergency department using initial triage information: An XGBoost algorithm analysis," JMIR Med. Inform., vol. 9, no. 9, Art. no. e30770, Sep. 2021, doi: 10.2196/3077.

[27] Y. Zheng, Z. Guo, Y. Zhang, J. Shang, L. Yu, P. Fu, Y. Liu, X. Li, H. Wang, L. Ren, W. Zhang, H. Hou, X. Tan and W. Wang, "Rapid triage for ischemic stroke: A machine learning-driven approach in the context of predictive, preventive and personalised medicine," EPMA J., vol. 13, pp. 285–298, May 2022, doi: 10.1007/s13167-022-00283-4.